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PES University, Bengaluru

(Established under Karnataka Act No. 16 of 2013)

Sample Paper for SMLR

Time: 3 Hrs	Answer All Questions	Max Marks: 100

Instructions

1. Answer all the questions.
2. Section A should be handwritten in the answer script provided.
3. Section B and C are coding questions which have to be answered in the system and uploaded.
4. Smartly use GridSearchCV as it might impact the system performance.

Section A (20 marks)			
1	a)	Explain frequency encoding and its role in handling categorical variables in Data Science.	5
	b)	Explain Box-cox transformation.	5
	c)	Write a note on Backward elimination method of feature selection algorithms.	5
	d)	How can be the problem of overfitting can be reduced in Linear regression? Explain the different variants to solve the over fitting problem in Linear regression.	5
Section B (40 Marks)			
2		Problem Statement: A firm aims to predict temperature (in Celsius) using various environmental and seasonal features. Using the provided dataset, perform appropriate Exploratory Data Analysis (EDA) to understand and preprocess the data, and then develop a machine learning model to predict temperature.	
	(i)	Read the given dataset and perform the following tasks: examine the dataset, identify the data types of all features, determine which features add little to no useful information, and visualize the relationship between Wind Speed (km/h) and Temperature.	4
	(ii)	Perform Exploratory Data Analysis (EDA) on the given dataset by checking for missing values, calculating the percentage of missing values in each column, and implementing an appropriate strategy to handle the missing data.	8
	(iii)	Create a new column as year, month, day and hour using the date column.	6
	(iv)	visualize the outliers in numeric columns using a boxplot and apply the IQR method to detect and eliminate the outliers.	8
	(v)	Generate a correlation plot using a color map to visualize feature relationships, and compute multicollinearity using VIF for the input features, sequentially removing features with high VIF values.	8
	(vi)	Remove all irrelevant features identified through the previous EDA steps and apply appropriate encoding techniques to transform the categorical variables.	6

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Section C (40 marks)			
3	(i)	Split the data into training and testing sets in a 80:20 ratio, build a Linear Regression model using the OLS method from the statsmodels package, and generate the model summary report.	8
	(ii)	Build a model using sklearn's least squares regression. Interpret the coefficients.	8
	(iii)	Determine the optimal subset of significant features using forward selection and backward elimination methods. Then, construct a model using the features obtained from forward selection and present the corresponding R-squared score.	8
	(iv)	Use 5-fold cross-validation to validate the model, print the RMSE scores for each fold, and explain whether the model is overfitting based on the results.	8
	(v)	Construct a Lasso regression model and perform hyperparameter tuning using Grid Search Cross-Validation to obtain the optimal alpha value.	8